

Facies architecture and sequence stratigraphy of an early post-rift fluvial succession, Aptian Barbalha Formation, Araripe Basin, northeastern Brazil



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ABSTRACT

The Barbalha Formation (Aptian) records deposition in a fluvial and lacustrine environment accumulated in an early post-rift sag basin. Characterization of the facies architecture and sequence stratigraphic framework of the alluvial succession was carried out through detailed description and interpretation of outcrops and cored wells. The development of depositional sequences in this unit reflects variation in the accommodation-to-sediment supply (A/S) ratio. Two depositional sequences, showing an overall fining-upward trend, are preserved within the succession. The sequences are bounded by regional subaerial unconformities formed during negative A/S ratio, and may be subdivided in Low-accommodation Systems Tracts (LAST) (positive A/S ratio close to zero) and High accommodation Systems Tracts (HAST) (A/S ratio between 0.5 and 1). Sequence 1, with a minimum thickness of 100 m, is characterized by amalgamated, multi-storey, braided fluvial channel sand bodies, defining a LAST. These are interlayered with crevasse splay and floodplain deposits toward the top, passing to open lacustrine deposits, defining a HAST. Sequence 2, with minimum thickness ranging from 50 to 90 m, overlies the organic-rich lacustrine deposits. At the base, this sequence is composed of amalgamated, multistorey braided fluvial channel sand bodies (LAST), similar to Sequence 1, overlain by well-drained floodplain with fixed fluvial channel deposits, interpreted as an anastomosed fluvial system, which are in turn capped by lacustrine deposits, both grouped in a HAST. Paleocurrent data on fluvial deposits of sequences 1 and 2 show a consistent paleoflow to the SE. Sedimentological evidence indicates humid to sub-humid climatic conditions during deposition of sequences 1 and 2. Accumulation of fluvial sequences 1 and 2 was mainly controlled by tectonics. Variation in A/S ratios must be related to tectonic subsidence and uplift of the basin.

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1. Introduction

Concepts of sequence stratigraphy have been applied to fluvial successions with relative success. The destruction or generation of new accommodation space in fluvial settings is controlled by variations in the fluvial profile (Schumm, 1993), related to changes in tectonics, climate and/or sea-level changes, which are often difficult to decipher from the 2D sedimentary record. The identification of systems tracts has been possible within alluvial systems based mainly on the relative proportion of fluvial channel sand bodies and fine overbank deposits. Intervals characterized by amalgamated fluvial channel sandbodies interlayered with rare overbank deposits indicate a succession accumulated with low rates of accommodation creation (Shanley and McCabe, 1991; Aitken and Flint, 1995; Olsen et al., 1995; Van Wagoner, 1995; Miall, 1996; Martinsen et al., 1999; Allen and Fielding, 2007; Scherer

et al., 2007, 2014; Fanti and Catuneanu, 2010). In turn, fluvial successions characterized by lenticular and isolated channels sandbodies, encased within fine-grained overbank deposits are formed during periods of higher rates of accommodation creation (Shanley and McCabe, 1991, 1994, 1994; Wright and Marriot, 1993; Aitken and Flint, 1995; Olsen et al., 1995; Van Wagoner, 1995; Martinsen et al., 1999; Cleveland et al., 2007; Fanti and Catuneanu, 2010).

Despite the consensus that the degree of amalgamation of fluvial channel sand bodies (channel/floodplain ratio) is the main criterion for the definition of systems tracts in fluvial successions, there is discussion about terminology and the number of systems tracts that make up an alluvial depositional sequence, and the mechanisms that control sediment accumulation (tectonics, climate and/or relative sea level). This demonstrates the need for more case studies that characterize the stratigraphic architecture and discuss the mechanisms controlling the accumulation of fluvial successions. This study aims to provide a well-constrained application of sequence stratigraphic concepts in alluvial successions, taking as a case study the Barbalha Formation (Aptian)

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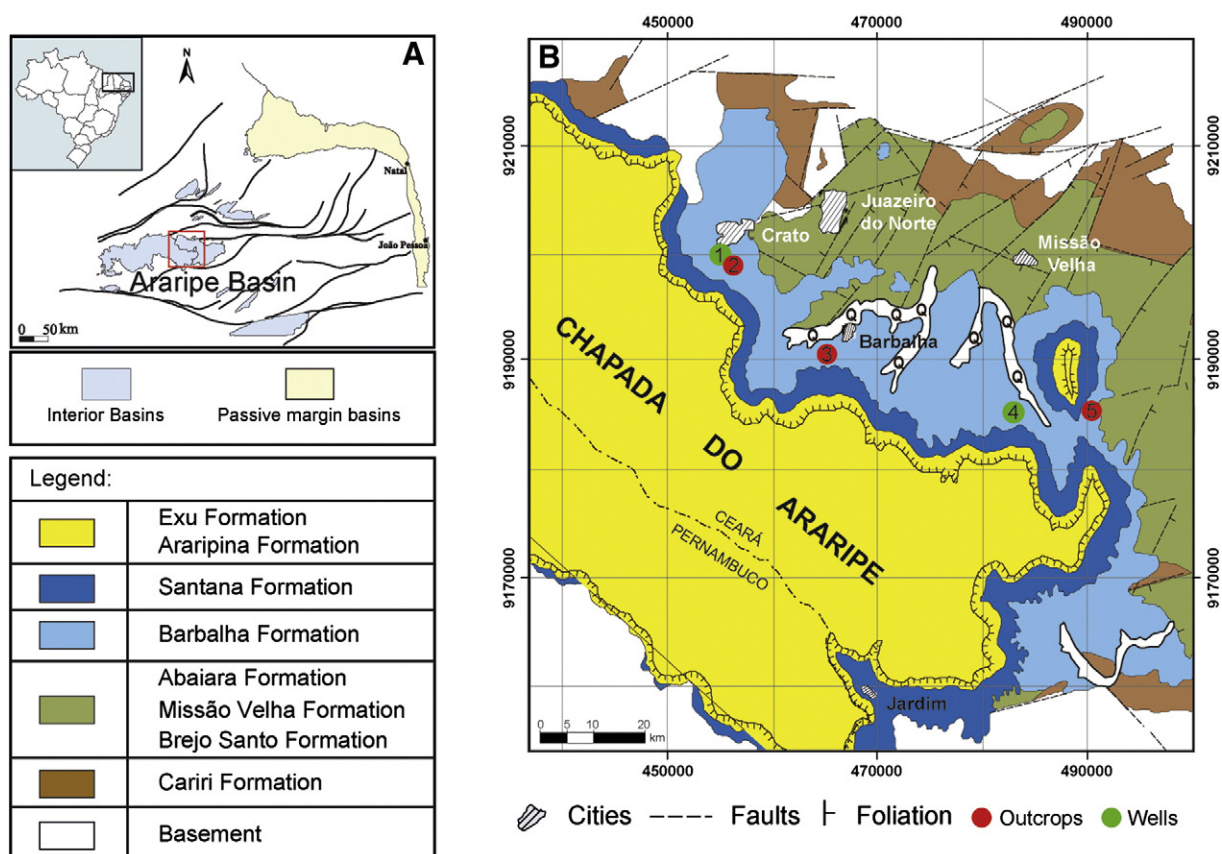


Fig. 1. (A) Location of the Araripe Basin in Northeastern Brazil. (B) Simplified geological map of the eastern part of the Araripe Basin, showing the location of five logged sections, which represent the study localities discussed in this paper.

of the Araripe Basin, northeastern Brazil. The goals are: (1) to analyze facies architecture and paleocurrent patterns; (2) to propose a sequence stratigraphic framework through the correlation of key surfaces, system

tracts and depositional sequences; and (3) to understand the stratigraphic evolution and to discuss the main controls on sedimentation and accumulation of the fluvial succession.

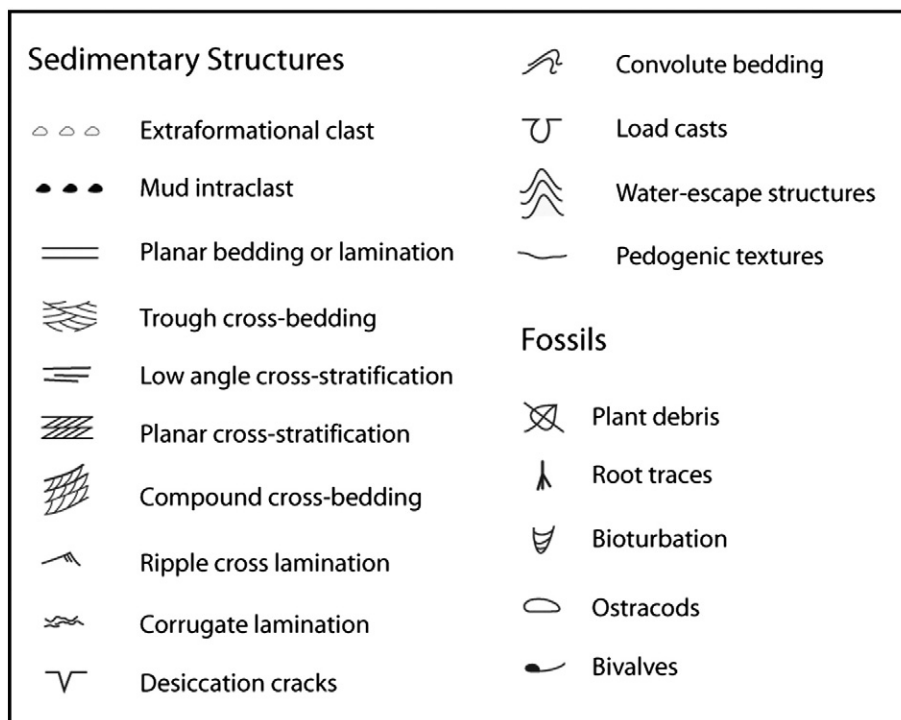


Fig. 2. Key to symbols used in graphic logs.

Table 1

Summary of lithofacies observed in the Barbalha Formation (based on Paula-Freitas (2010) and data collected in this work).

Facies	Description	Interpretation
Gm	Massive or crudely-stratified, clast-supported conglomerates; coarse sandstone matrix; mud intraclasts (<30 cm) are dominant, rare granite and quartz clasts (<5 cm) are also present; 10 to 30 cm thick beds.	Bedload deposition as diffuse gravel sheets (Hein and Walker, 1977) or lags deposits by high-magnitude flood flows (Miall, 1977; Nemec and Postma, 1993).
Gt	Sandy conglomerates; granule to pebble clasts; extraformational clasts of granite and quartz; medium to coarse sandstone matrix, trough cross-stratified; 10 to 40 cm thick sets.	3D gravel dunes (Rust, 1978; Todd, 1996).
Sm	Fine- to very coarse-grained sandstones; moderately- to well-sorted; massive; burrows in some fine sandstones; 20 to 80 cm thick beds.	Rapid deposition of hyperconcentrated flows, fluidization or intensive bioturbation (Miall, 1978, 1996).
St	Medium-grained to granular sandstones; poorly- to well-sorted; extraformational clasts of quartzite, granite and quartz (<3 cm) and mud intraclasts (<11 cm) at the base of sets, or parallel to stratification; trough cross-stratification with 20 cm to 2 m thick sets; sometimes burrows in the topmost part of, or throughout, the bed.	3D subaqueous sandy dunes (lower flow regime); later modified by bioturbation (Allen, 1963; Harms et al., 1982; Todd, 1996; Collinson et al., 2006)
Sp	Medium-grained to granular sandstones; poorly to well-sorted; rare extraformational granules and pebbles of granite and quartz at the base of sets, or parallel to stratification; planar cross-stratification; 10 to 50 cm thick sets.	2D subaqueous sandy dunes (lower flow regime); later modified by bioturbation (Allen, 1963; Harms et al., 1982; Todd, 1989; Collinson et al., 2006)
Ss	Medium to coarse-grained sandstones; moderately sorted; sigmoidal cross-bedding; 20 to 30 cm thick sets.	Lower- to upper-flow regime transitional bedform (Wizevich, 1992).
Sl	Medium-grained to granular sandstones; poorly- to well-sorted; granule and pebble of granite and quartz (<2 cm) parallel to stratification; low-angle cross-stratification; 20 cm to 1.90 m thick beds.	Washed-out dunes and humpack dunes (transition between subcritical and supercritical flows) (Harms et al., 1982; Bridge and Best, 1988).
Sh	Very fine- to fine-grained sandstones; moderately- to well-sorted; horizontal lamination; 10 to 40 cm thick beds.	Planar-bedded deposits originated via upper flow regime (Miall, 1977; Best and Bridge, 1992).
Sr	Fine to medium-grained, micaceous sandstones; ripple cross-lamination, set thickness a few cm, forming up to 1.5 m thick cosets; supercritical to subcritical climbing angle	2D- or 3D-ripples (lower flow regime) (Allen, 1963; Miall, 1977).
Fm _r	Mudstones to very fine-grained sandstones; massive; sometimes fissile in weathered surfaces; reddish to reddish brown; diffuse horizons of color variations; mottle and block structures; root traces; slickensides; 20 cm to 3.5 m thick beds.	Suspension settling on floodplains; later modified by desiccation or pedogenetic processes; post-depositional reddening under oxidizing conditions (Miall, 1977; 1990; Foix et al., 2013).
Fm _g	Mudstones to very fine-grained sandstones; massive; medium gray, brownish gray and greenish gray; sometimes burrowed; with ostracod, conchostracan and gastropod fossils; 20 cm to 8 m thick beds.	Suspension settling from weak currents or standing water; lack of lamination due to (i) flocculation of clay suspension or (ii) loss of lamination associated intensive bioturbation; post-depositional graying under reducing conditions (Miall, 1977; Foix et al., 2013).

Table 1 (continued)

Facies	Description	Interpretation
Fl _g	Mudstones; thin parallel lamination; medium gray, brownish gray and greenish gray; with ostracod, conchostracan and gastropod fossils; 20 cm to 6 m thick beds.	Suspension settling dominantly from standing water; post-depositional graying under reducing conditions (Turner, 1980; Jo and Cough, 2001).
Fl _{org}	Organic-rich mudstone; laminated; dark gray to black; interlayered with millimetric to centimetric (<2 cm) microbial carbonate laminae or nodules and/or ostracod-rich levels; lack of bioturbation.	Suspension settling of sediments under anoxic or poorly-oxidized conditions (Chakraborty and Sarkar, 2005; Foix et al., 2013)

2. Geological setting

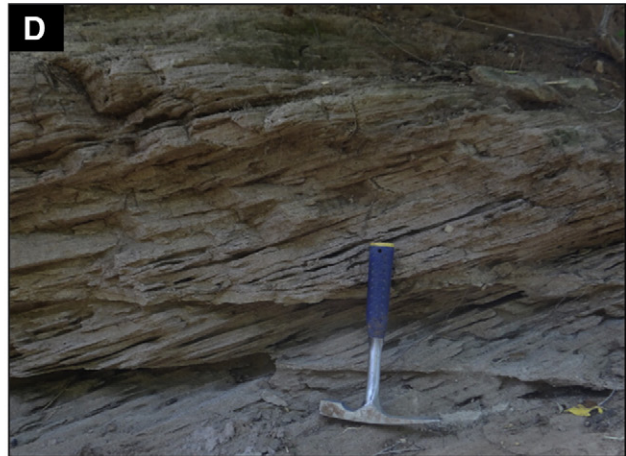
The Araripe Basin covers more than 9000 km², and is the most extensive interior basin in northeastern Brazil (Fig. 1). The Araripe Basin was formed by extensional processes related to the fragmentation of Gondwana and opening of the eastern Brazilian continental margin during the Cretaceous (Guignone et al., 1986; Assine, 2007). This polyphase basin reflects different stages of subsidence related to three main phases (Ponte and Ponte Filho, 1996): (a) the Pre-Rift phase, characterized by regional subsidence due to viscous–elastic lithospheric stretching; b) the Rift phase, with accentuated mechanical subsidence, forming graben and/or half-graben systems; and c) the Post-Rift phase, characterized by the predominance of thermal subsidence, that included the Aptian/Albian succession studied in this paper.

According to the studies by Ponte and Appi (1990), Assine (1990, 1992); Ponte and Ponte Filho (1996), the Araripe Basin may be subdivided into sequences bound by regional unconformities that reflect distinct tectonic stages in the basin. Assine (2007) integrated these different proposals, identifying five large unconformity-bound units: (1) the Paleozoic Sequence, represented by alluvial sedimentation of the Cariri Formation, interpreted as residual deposits of a large intracratonic basin; (2) the Pre-Rift Supersequence (Neojurassic), corresponding to the Brejo Santo and Missão Velha formations; (3) the Rift Supersequence (Neocomian), equivalent to the Abaiara Formation; (4) the Post-Rift Sequence I, comprising the Barbalha and Santana formations; and (5) the Post-Rift Sequence II, equivalent to the Araripina and Exu formations.

The Barbalha Formation, in the basal section of the Post-rift sequence I (equivalent to the Rio da Batateira Formation, proposed by Ponte and Appi (1990)), is composed of fluvial and lacustrine deposits, with an average thickness of about 200 m (Assine, 1992, 2007). This unit can be subdivided into two large fining-upward cycles, starting with sandy conglomerates to coarse-grained sandstones, which pass upwards to fine-grained sandstones and mudstones (Assine, 1992, 2007; Chagas et al., 2007). The top of the first cycle is characterized by an interval comprising organic, black shales interlayered with carbonate-rich laminae, named the “Batateira layers” (Hashimoto et al., 1987; Assine, 1992, 2006, 2007) or the Fundação Member (Rios-Netto et al., 2012). Coprolites, ostracods, fish remains and plant fragments are abundant. Based on the fossil content, an upper Aptian age was assigned for this unit (Farina, 1974; Lima and Perinotto, 1984; Hashimoto et al., 1987; Regali, 2001; Coimbra et al., 2002). Paula-Freitas (2010) subdivided the Aptian–Albian (the interval including the Barbalha Formation) in four depositional sequences, based on outcrops and cored wells. However, these sequences are difficult to identify, since the subaerial unconformities are not easily correlated between different stratigraphic sections.

3. Study area and methods

The Barbalha Formation crops out along the eastern side of the Araripe Basin (Fig. 1) covering more than 10,000 km². Data presented



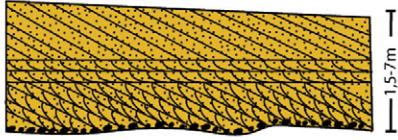
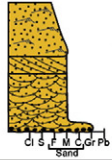
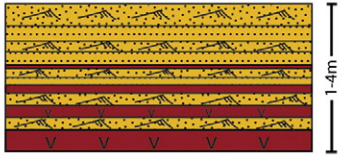
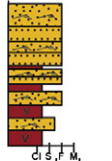
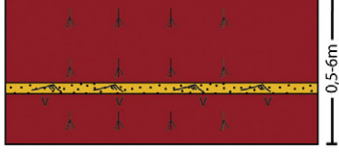
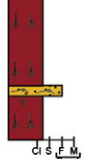
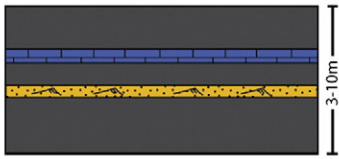
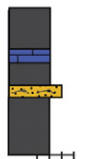
Facies association	Facies architecture and geometry	Vertical log	Description	Lithofacies	Interpretation
Multistorey, amalgamated fluvial channel			Fine- to very coarse-grained sandstones, 1.5 to 7 m thick; basal erosional surface with up to 1.5 m of relief; composed of frontal or lateral accretion macroforms and sand bedform.	Gm, Gt, St, Sp, Si, Sh, Sr	Low-sinuosity, braided fluvial channel belt.
Single storey, fluvial channel			Fine- to coarse-grained sandstones, isolated ribbon sandstone bodies; 1 to 3 m thick; sometimes forming a fining-upward succession.	Gm, St, Sp, Sr	Fixed, anastomosing fluvial channel.
Crevasse Splay			Very-fine to fine sandstones interlayered with mudstone layers; tabular sheet-like bodies, 1 to 4 m thick; desiccation cracks are common; coarsening and thickening-upward succession.	Sh, Si, Sr, St, Fmr	Crevasse splays.
Floodplain			Massive red mudstones; abundant paleosol development; sheet-like or irregular tabular bodies; 0.5-4 m thick.	Fmr, Fmg, Flg, Sr	Well-drained to poor drained (more rare) floodplain.
Lacustrine			Black, organic-rich mudstones, rarely interlayered with massive or ripple cross-laminated fine-grained sandstones and microbial carbonates; common bivalve and ostracode fossils, sometimes forming bioclastic laminae; tabular package, 3-10m thick.	Florg, Flg, Sr, Sm	Anoxic, perennial, shallow lake.

Fig. 4. Description and interpretation of the facies association observed in the Barbalha Formation.

in this study correspond to Cariri sub-basin. The data for this paper were collected through detailed description and interpretation of the stratigraphic sections logged in exposures along rivers in three localities and two cored well-logs (Fig. 1B). High-resolution sedimentary logs were measured in outcrops and wells to define the main facies and facies association of the studied interval (Fig. 2). Architectural panels were constructed from the photomontages of the selected outcrops, in which two-dimensional (2D) geometries of the deposits were defined. Facies were classified mainly on the basis of grain-size and sedimentary structures, following the schemes of Miall (1978). The facies were grouped in facies associations, which correspond to a group of facies genetically related to one another, representing subenvironment within a depositional system (Collinson, 1996; Dalrymple, 2010). A detailed proposal of facies associations of the Barbalha Formation has not been done before. Previous sedimentological studies in the Barbalha Formation (e.g., Chagas, 2006; Paula-Freitas, 2010) made a general reconstruction of fluvial depositional system without detailed description, documentation and interpretation of the constituent facies associations. Paleocurrent orientations were measured from unidirectional sedimentary structures, mainly trough and planar cross-bedding. Stratigraphic key surfaces and system tracts were identified and correlated between

different sedimentary logs, allowing the individualization of different depositional sequences.

4. Lithofacies

The Barbalha Formation is composed of dominantly quartzose sandstones, mudstones and subordinate conglomerates. Texturally, the sandstones are very fine- to coarse-grained (mostly medium-grained), moderately-sorted, with subangular to rounded grains. Quartz pebbles are common, especially near the base of the cross-bedded sets. The mudstones are commonly massive, or laminated. Bioturbation and body fossils are found in fine- to very fine-grained sandstones and mudstones. The conglomerates are sandy, massive or stratified, with subangular to subrounded granules to pebbles, dominantly quartz and, subordinately, feldspar and granitic fragments. Thirteen lithofacies are recognized within the Barbalha Formation, summarized in Table 1 and illustrated in Fig. 3. The facies used in this paper are similar to those proposed by Paula-Freitas (2010), with an agreement in the description and interpretation.

Fig. 3. Representative lithofacies of the Barbalha Formation. (A) Massive, intraformational conglomerate (Gm); (B) Trough cross-bedded pebbly, sandy conglomerate (Gt); (C) Trough cross-bedded sandstone (St); (D) Planar cross-bedded sandstone (Sp); (E) Ripple cross-laminated sandstone (Sr); (F) Reddish, massive mudstones (Fm); (G) Grayish, laminated mudstone (Flg); (H) Organic, black shale with carbonate laminae and nodules (Florg).

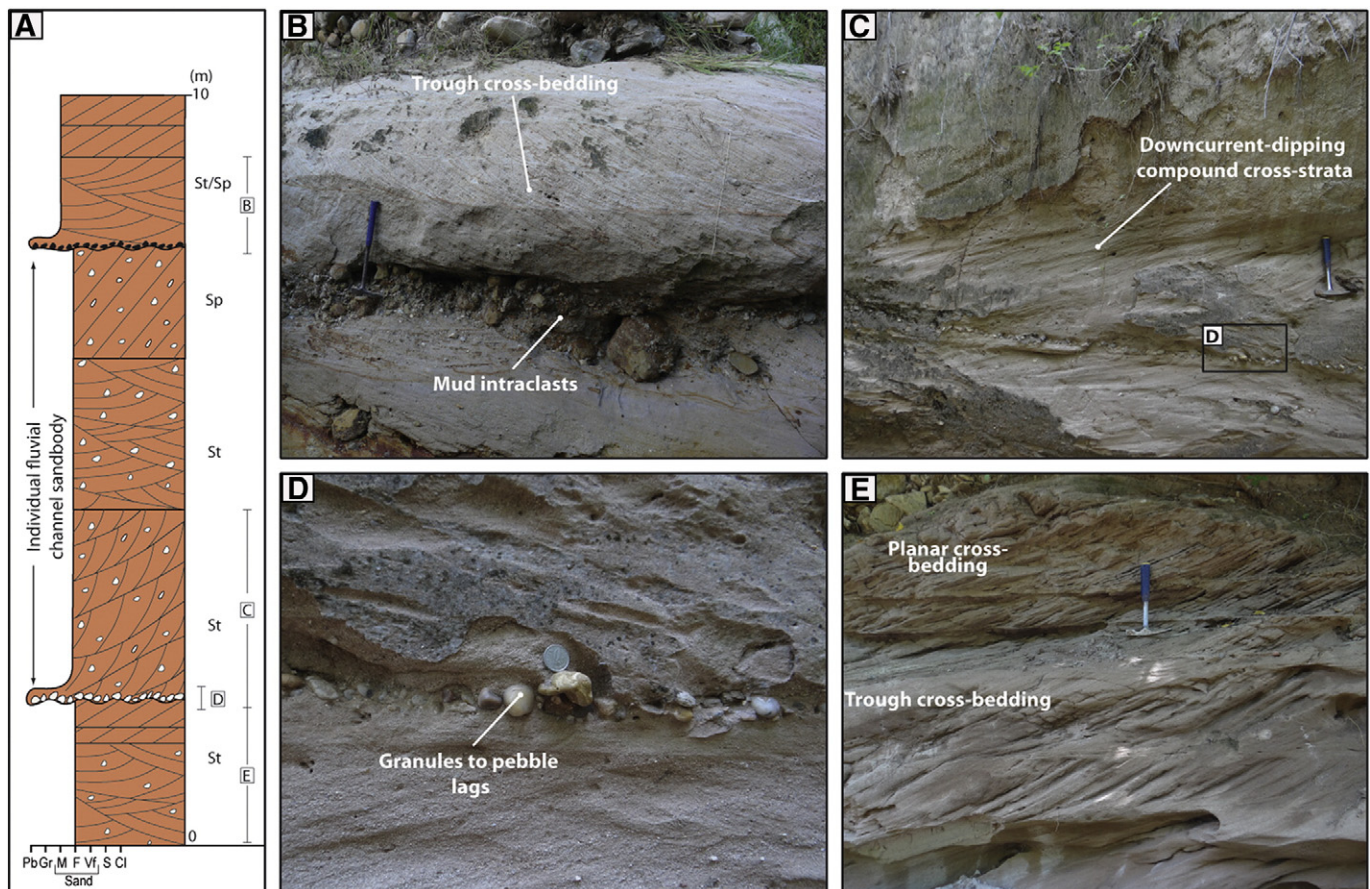


Fig. 5. (A) Typical vertical log of multistorey, fluvial channel sand bodies facies association. (B) Sand bodies with trough cross-bedded (St) bounded by erosional surface covered by massive, intraformational conglomerate (Gm). (C) Sandstones showing downcurrent dipping compound cross-strata. (D) Detail of (C), displaying quartz granules and pebbles forming a lag at the base of the sand body channel. (E) Coset of the cross-strata, characterized by trough (St) and planar (Sp) cross-bedded sets, where the individual sets are bounded by horizontal surfaces, defining an architectural element with horizontal, cross-stratified sets. See Fig. 2 for explanation of symbols used in the sedimentological logs.

5. Facies associations

The Barbalha Formation comprises five genetic facies associations: (1) Multistorey, fluvial channel sand bodies; (2) Single-storey, fluvial channel sand bodies; (3) Crevasse Splay; (4) Floodplain; and (5) Lacustrine (Fig. 4).

5.1. Multistorey, fluvial channel sand bodies facies association

5.1.1. Description

This facies association consist of several sandstone bodies up to 7 m thick, composed mainly by fine- to coarse-grained sandstones with planar (Sp) and trough cross-bedding (St) and, less commonly, sigmoidal cross-bedding (Ss), horizontal lamination (Sh), low-angle cross-bedding (Sl) and ripple cross-lamination (Sr). The sandstone bodies internally show a poorly-defined fining-upward trend, and they are bounded by flat to concave-up surfaces (Fig. 5). In places, massive, intraformational conglomerates (Gm) or trough cross-bedded, sandy conglomerates (Gt) cover the basal erosive surface of the sandstone bodies. In outcrops, the sandstone bodies show sheet geometry, extending laterally for hundreds of meters. The sandstone bodies are multistorey, composed of different architectural elements, identified on the basis of lithofacies assemblage, geometry and orientation of the boundary surfaces (Fig. 6):

Simple, large-scale cross-strata: These comprise solitary sets of cross-strata that are 1.0–2.5 m thick, extending laterally for more than 25 m. Foresets are planar to sigmoidal in shape, with uniform dip directions and dip-angles throughout the sets. Occasionally, closely spaced (20–30 cm) internal erosional surfaces, can be observed separating

foresets with similar orientation (Figs. 7, 8). They display a foreset-truncating, near-horizontal erosion surface in the upper part that turns downwards along the cross-strata to become concordant and non-erosional to the base of sets, corresponding to a “planar-convex minor contact” of Allen (1983).

Downcurrent-accretion compound strata: These comprise grouped sets of trough (St) and planar (Sp) cross-strata, forming storeys 1.5–3.0 m thick. Sets have a mean thickness of 20 cm and are bounded by flat surfaces dipping at low angles (5 to 15°), in the same direction as the cross-bedding (Figs. 7, 8).

Lateral-accretion compound strata: These consist of cosets (1–2 m thick) composed of planar (Sp) and trough cross-strata (St), 10–20 cm thick, which are bounded by low-angle, inclined surfaces (dipping less than 12°), highly oblique to perpendicular to the direction of the cross-strata dip (Fig. 9).

Horizontal, cross-stratified sets: These consist of stacked sets (cosets) of planar (Sp), trough (St) and rare sigmoidal (Ss) cross-bedding. Individual cross-stratified sets are 20–30 cm thick, forming cosets 1–3 m thick and tens of meters in lateral extent. The sets are limited by horizontal or gently inclined surfaces, dipping with opposite orientation to the cross-strata direction.

5.1.2. Interpretation

The occurrence of amalgamated sand bodies bounded by erosive surfaces, composed mainly of fine- to coarse-grained sandstones, moderately to poorly-sorted, arranged in stacked sets of unidirectionally-oriented, decimeter-scale planar and trough cross-strata, suggests deposition in fluvial channels according to the interpretation of Chagas (2006); Paula-Freitas (2010). However, the interpretation in

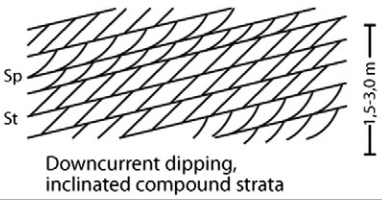
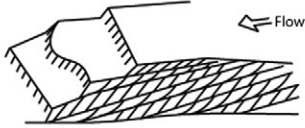
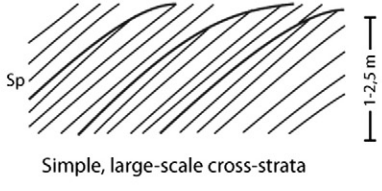
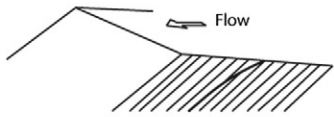
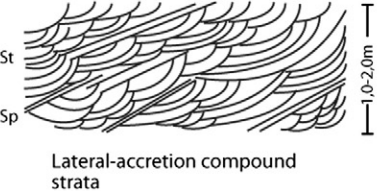
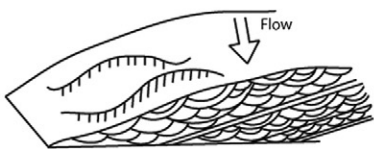
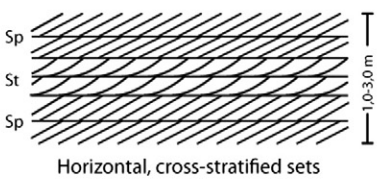
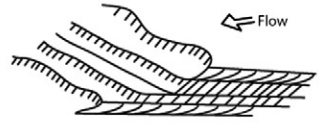
Architectural element	Description	Interpretation
 Downcurrent dipping, inclined compound strata	Medium- to coarse-grained sandstones, organized in cosets of planar and trough cross-stratification. Individual sets are bounded by low-angle, inclined surfaces, dipping in the same direction of the intraset cross-strata.	Downstream accretion of middle-channel bars, with superimposed dunes. 
 Simple, large-scale cross-strata	Medium- to coarse-grained sandstones, organized in solitary sets of large-scale planar cross-stratification (0.8 to 1.5 m thick). Internal sigmoidal erosional surfaces.	Downstream migration of sandy bars with well-developed slipfaces, such as transverse or alternate bars. 
 Lateral-accretion compound strata	Medium- to coarse-grained sandstones with large-scale compound cross-strata, dipping at high-angles or perpendicular to the direction of the cross-strata dip.	Lateral accretion in middle channel bars. 
 Horizontal, cross-stratified sets	Fine to coarse sandstones arranged in stacked sets of trough or planar cross bedding, bounded by horizontal surfaces. Individual sets are 15–40 cm thick.	Vertical accretion and downstream migration of dunes bedforms above channel floor or bar top. 

Fig. 6. Summary of architectural elements and constituent facies observed in the multistorey fluvial channel sand bodies facies association.

these papers are generic, not reconstructions of the fluvial architecture. The detailed facies analysis presented in this work allows a more precise reconstruction of the fluvial channel anatomy. The simple, large-scale cross strata are interpreted as formed by downstream migration of sand bars with well-developed slipfaces, such as transverse bars, alternate bars or tributary mouth bars (Wizevich, 1992; Bridge, 1993; Jo and Chough, 2001). Erosional contacts that occur inside of simple, large-scale cross-strata (planar-convex minor contact *sensu* Allen, 1983) represent reactivation surfaces. This kind of surface reflects local variations in current strength and/or sediment supply that cause a bedform disequilibrium making the top and cross-sets of the bar advance at different rates (Allen, 1983; Hjellbakk, 1997). The regular spacing between the reactivation surfaces can be related to periodic (seasonal?) oscillations in river discharge. Downcurrent-dipping, inclined compound strata represent the downstream accretion of compound sand bars with superimposed dunes. This element is similar in internal architecture and facies assemblage mid-channel bars (Allen, 1983; Haszeldine, 1983; Wizevich, 1992; Miall, 1996). The large-scale inclined strata dipping at high angles with or perpendicular to local paleocurrent represent the lateral accretion of compound bars, probably associated with mid-channel bars (Miall, 1996; Jo and Chough, 2001). Horizontal, cross-stratified sets can be interpreted as fields or trains of individual bedforms that accumulated predominantly by vertical aggradation associated with downstream migration and climbing of subaqueous dunes, comparable to sand bedforms of Miall (1985, 1996). These fields of 3D dunes probably represent deposits accumulated in the deeper portion of fluvial channels, developed between mid-channel macroforms (Bristow, 1987; Miall, 1994; Jo and Chough, 2001). However, the small-

scale cross-bedding cosets can represent small dunes that occur in shallow areas of active channels, particularly on bar tops (Miall, 1996). The multistorey, sheet-like geometry of the sand bodies, prevailing coarse-grained nature of the deposits, low dispersion of paleocurrents, lack of well-defined fining-upward cycles, and the abundant occurrence of simple to compound downstream accretion macroforms collectively suggest a low-sinuosity, braided fluvial channel belt pattern. The lateral and vertical association of the different architectural elements is indicative of a number of bars and channels in a braided river (Jo and Chough, 2001; Jo, 2003). The sheet-like nature of the sandstone bodies indicates highly mobile fluvial channels (Friend, 1983). The thickness between 1–3 m of the bar deposits indicates that the maximum bankfull channel depth of the braided channels was 6 m, according the assumption that height of the macroforms is between one-half and the total channel depth during bankfull stage (Bristow, 1987). The abundance of cross-bedding suggests a relatively constant discharge regime, indicating perennial fluvial channels (Miall, 1996; Allen et al., 2013).

5.2. Single-storey, fluvial channel sand bodies facies association

5.2.1. Description

This facies association is made up of lenticular sand bodies, up to 3 m thick, which extend laterally for a few tens of meters, encased in fine-grained deposits of the floodplain and crevasse splay facies associations (Fig. 10A, B). The sand bodies are bounded by a low- to high-relief basal erosion surface, and they are composed of fine- to coarse-grained sandstones with trough cross-bedding

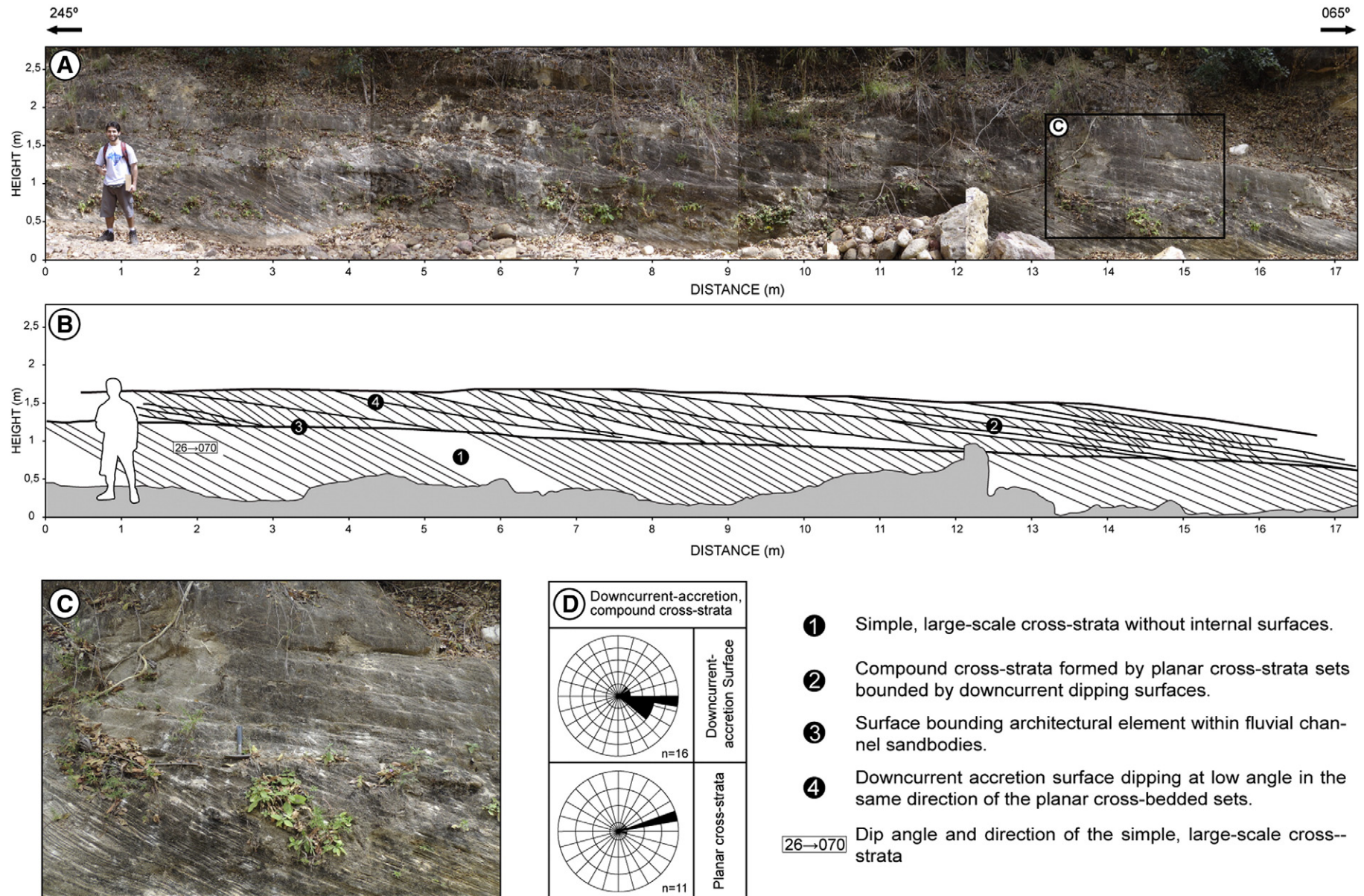


Fig. 7. Photomosaic (A) and interpreted outcrop panel (B) showing the different architectural elements of multistorey and multilateral, fluvial channel sand bodies facies association. (C) Close-up of (B) showing the contact between simple, large scale cross-strata and compound cross-strata. (D) Rose diagrams of the accretion surfaces and planar cross-strata, showing a similarity of the dip direction that indicates downcurrent accretion macroforms.

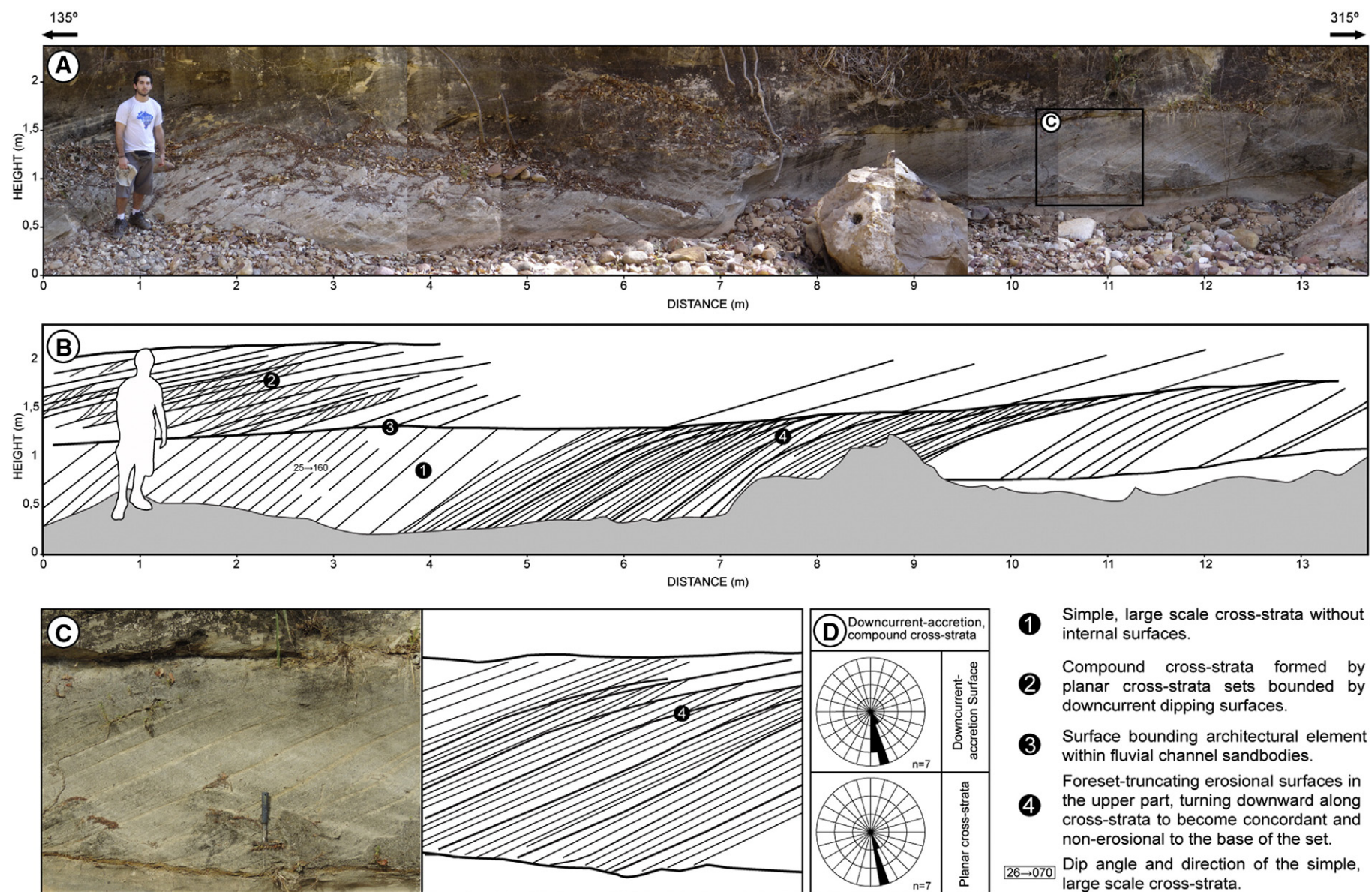


Fig. 8. Photomosaic (A) and interpreted outcrop panel (B) showing different architectural elements of multistorey, fluvial channel sand bodies facies association. (C) Close-up of (B) showing reactivation surfaces. (D) Rose diagrams of the accretion surfaces and planar cross-strata, showing a similarity of the dip direction that indicates downcurrent accretion macroforms.

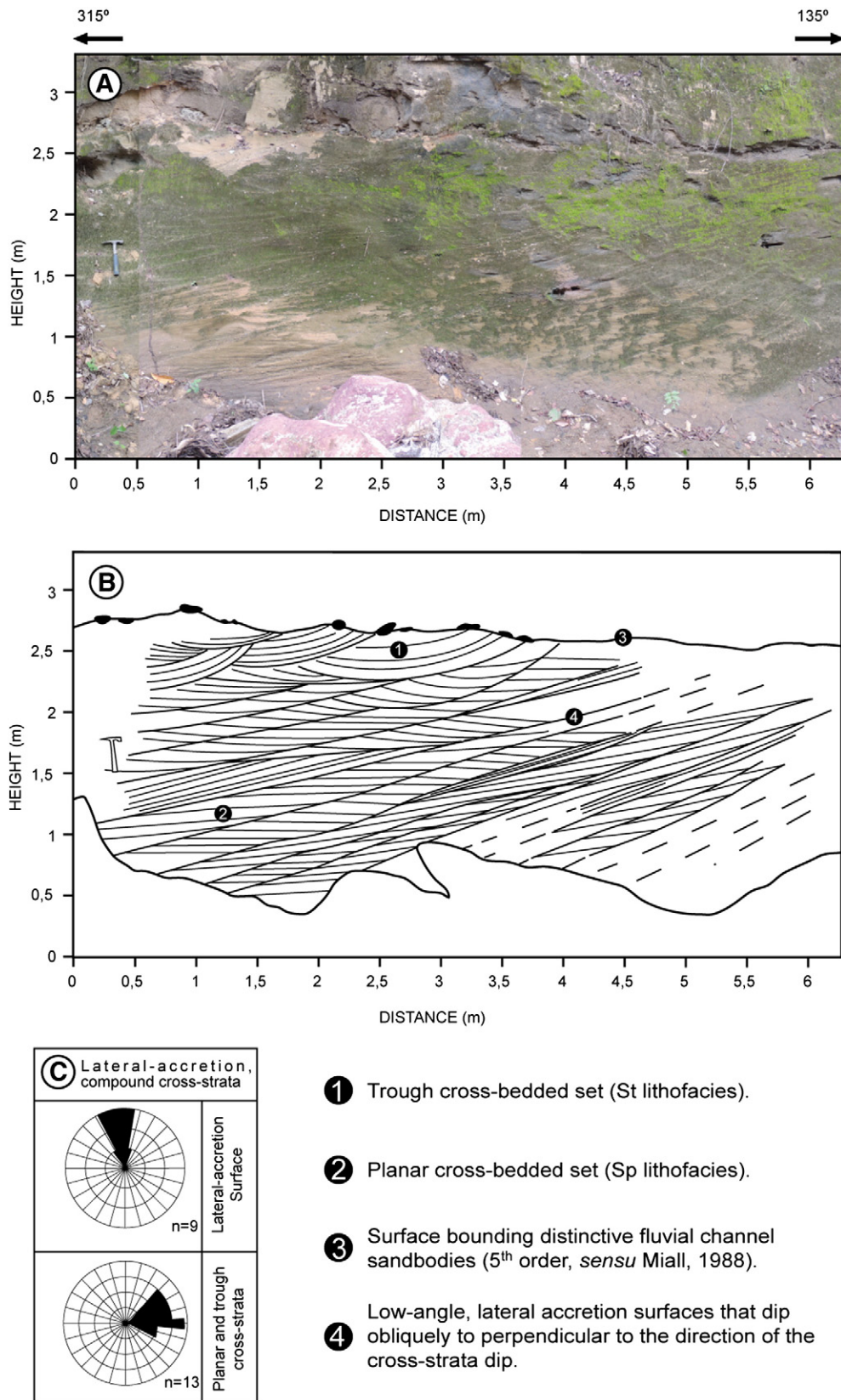


Fig. 9. Photomosaic (A) and interpreted outcrop panel (B) showing lateral-accretion compound strata. The planar and trough cross-strata are bounded by low-angle, inclined surfaces that dip in high angle to perpendicular to the direction of the cross-strata dip (C).

(St), ripple cross-lamination (Sr) and, rarely, low-angle cross stratification (Sl) (Fig. 10A, C). Internally, they are characterized by the absence of lateral accretion surfaces, and sometimes they show a well-defined fining-upward succession (Fig. 10A).

5.2.2. Interpretation

The presence of lenticular sand bodies bound by basal erosional surfaces, composed of decimetric cross strata with unidirectional paleocurrents allow the interpretation of this facies association as fluvial

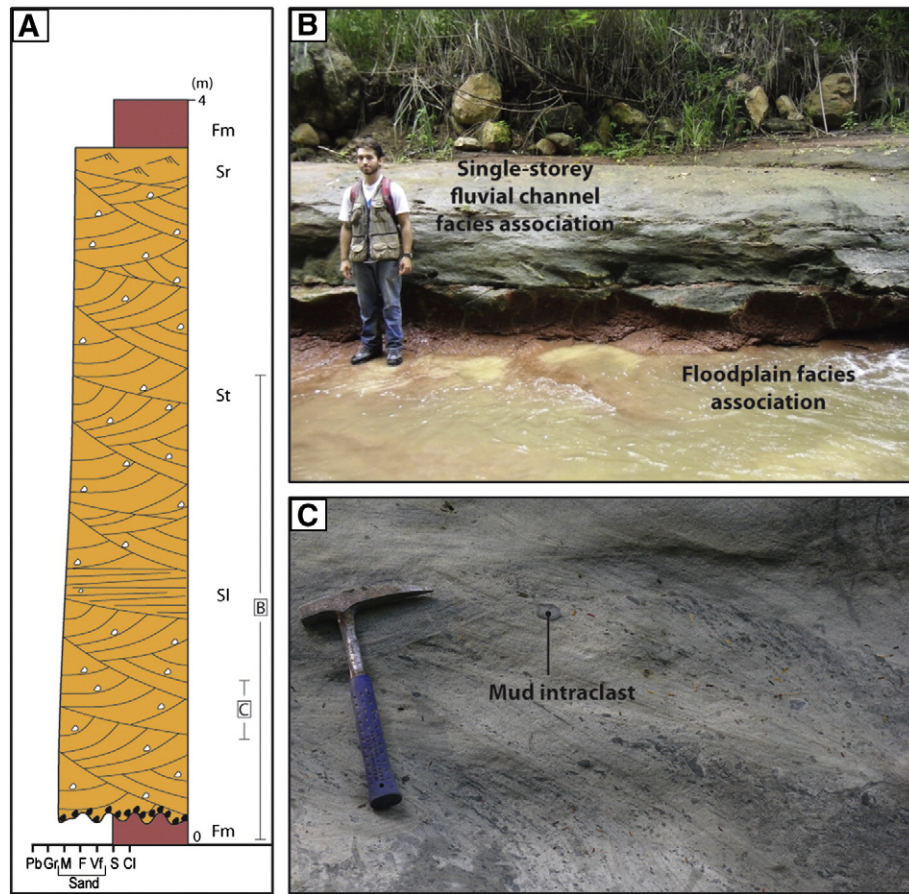


Fig. 10. (A) Typical vertical log of single-storey, fluvial channel sand bodies facies association. (B) Single-storey fluvial channel encased in floodplain deposits. (C) Detail of trough cross-bedded sets (St) with abundant intraclasts. See Fig. 2 for explanation of symbols used in the sedimentological logs.

channel deposits. The absence of lateral accretion elements suggests laterally stable, low sinuosity channels (Miall, 1996). As a result, single-story, lenticular sandstone beds are interpreted as the deposits of fixed channels, intimately associated with the floodplain deposits within which they are encased (Paredes et al., 2007). The fining-upward cycles indicate a progressive decrease in flow velocity during the fill of fluvial channels. This facies association was interpreted as meandering fluvial channel by Paula-Freitas (2010). However, the absence of lateral accretion surfaces associated with point bars, and the dominance of aggradational bedforms filling the sand bodies suggest fixed channels, instead of lateral migrating, meandering rivers.

5.3. Crevasse splay facies association

5.3.1. Description

This facies association is characterized by heterolithic bedding forming sheet bodies, 1–4 m thick and laterally extensive for tens of meters (Fig. 11A, B). The sheet bodies are characterized by rhythmic interlayers of massive, brown-to-reddish (Fm) or gray (Fmg) mudstones, and fine- to medium-grained sandstones (Fig. 11B, C). The sandstone beds are massive (Sm) or composed of horizontal (Sh) to low-angle cross-stratification (SI) and ripples cross-lamination (Sr). The reddish mudstones sometimes show mud cracks (Fig. 11C), and the sandstones are commonly bioturbated (Fig. 11D). Some sandstone beds show vertical distribution of facies, with Sh or SI lithofacies at the base, overlain by Sr lithofacies (Fig. 11E). The heterolithic packages are characterized by variation in the proportion of sandstone and mudstone. Some of them are dominated by medium to fine sandstone interlayers with millimeter- to centimeter-thick mudstones, while others are characterized by the dominance of muddy heterolithics. Coarsening- and

thickening-upward cycles, 2–4 m thick, are locally present, marked by a progressive increase of sandstone/mudstone ratio toward the top. This facies association commonly overlies well-drained floodplain or lacustrine overbank deposits, being the contact either gradual or sharp.

5.3.2. Interpretation

This facies association is interpreted as crevasse splays in the overbank area. The abundant ripple cross-lamination and horizontal to low-angle lamination suggests shallow flows, deposited by flow expansion and loss of flow competence as discharge leaves the confines of the channel (Miall, 1996). Massive sandstones represent hyperconcentrated flows or severely bioturbated sediments (Miall, 1996). The rhythmic interlayering between fine sandstones and mudstones represents the alternation of traction and suspension processes associated with different events of lateral spilling of a fluvial channel. The common vertical distribution in the sandstone beds of horizontally stratified to ripple cross-stratified represents the evolution of single depositional events, characterized by a progressive decrease in flow velocity (waning flow). Interlayered mudstones are interpreted as late-stage suspension fallout following initial flooding. The presence of mud cracks suggest periods of non-deposition and desiccation between flooding events. Coarsening-upward successions are indicative of successive flows arriving on a low-energy, well-drained floodplain or lacustrine overbank, marking the progradation of crevasse splays.

5.4. Floodplain facies association

5.4.1. Description

This facies association has a variable thickness of 50 cm to 6 m, and is laterally extensive for hundreds of meters (Fig. 12A, D). These deposits

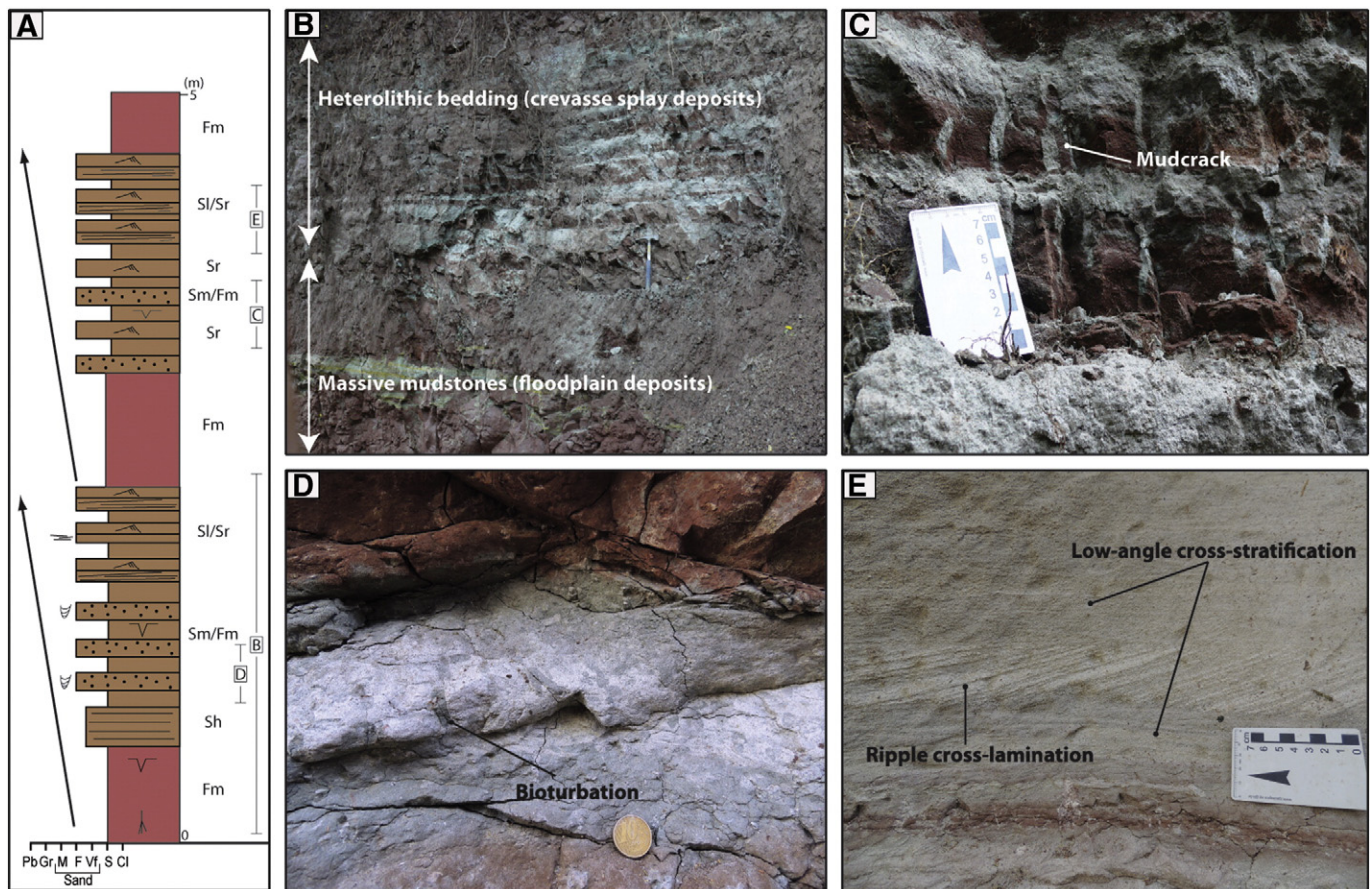


Fig. 11. (A) Typical vertical log of crevasse splay facies association. (B) Coarsening- and thickening-upward succession marked by red, massive mudstones at the base (floodplain facies association) passing upward to heterolithic bedding. (C) Detail of heterolithic bedding, characterized by alternation between massive sandstones (Sm) and reddish, massive mudstones with mud cracks (Fmr). (D) Red to greenish mudstones with bioturbation. (E) Sandstones with alternation of low-angle cross-stratification (Sl) and ripple cross-lamination (Sr), suggesting variation in flow velocity during deposition. See Fig. 2 for explanation of symbols used in the sedimentological logs.

are composed of mainly reddish, massive mudstones (Fmr) and, subordinately, laminated to massive gray mudstone (Fmg, Flg) (Fig. 12A). The reddish, massive mudstones (Fmr) have red-gray mottles, slickensides along fracture planes, carbonate globules, root traces and platy and blocky peds in some beds (Fig. 12B–D). Trace fossils are common in the form of vertical and passively filled tubes. This facies association occurs interlayered with a single-storey, fluvial channel facies association and, less commonly, with a multistorey, amalgamated fluvial channel facies association.

5.4.2. Interpretation

The fine deposits are interpreted to represent floodplain deposition. The occurrence of reddish mudstones associated with the presence of root traces, bioturbation, mottling, slickensides and platy and blocky peds, indicate well drained floodplains, which underwent periods of non-deposition and subsequent oxidation and development of paleosols (Bown and Kraus, 1987; Kraus, 1999; Therrien, 2006; Cleveland et al., 2007; Retallack, 2008). Surfaces with slickensides are interpreted to have formed by shrinking and swelling of clays, associated with episodic water infiltration and evaporation (Retallack, 1994). The carbonate nodules indicate the presence of alkaline solutes, precipitated due to fluctuations in the groundwater table (Retallack, 1994; Paredes et al., 2007). However, the sparse occurrence of the carbonate nodules suggests that partial leaching may have occurred as a result of well-drained conditions throughout the floodplain, and/or limited amounts of carbonate in the source area (Mack et al., 1993; Retallack, 2008). Mottling is formed by periodic waterlogging and reflects

localized decomposition and reduction of organic material (Retallack, 1988, 1994). The blocky and platy peds are interpreted as argillans formed in clay-rich paleosols by clay translocation or illuviation (Sigleo and Reinhardt, 1988). The laminated gray mudstone is interpreted as overbank deposition on a waterlogged floodplain with reducing conditions (Jo and Chough, 2001).

5.5. Lacustrine facies association

5.5.1. Description

This facies association consist mostly of gray mudstones, massive (Fmg) or laminated (Flg) and organic black shales (Florg), arranged in packages 3–10 m thick (Fig. 13A–C), traceable laterally for distances greater than 20 km. Fossil content is abundant and includes invertebrate (ostracods, bivalves, conchostracans) and plant remains (leaves and woods). Millimetric to centimetric (<2 cm) microbial carbonate laminae or nodules and/or ostracod-rich levels may be interlayered with organic, black shales (Fig. 13D). This facies association also comprises fine- to very fine-grained sandstones, 10 to 40 cm thick, massive (Sm) or with faint ripple cross-lamination (Sr). The sandstones are characterized by various types of feeding burrows, varying in diameters from a few mm to 2 cm.

5.5.2. Interpretation

The mud-dominated nature of this association is indicative of deposition in a quiet-water environment, with low sedimentation rates. The invertebrate fossils support a fresh- to brackish water environment.

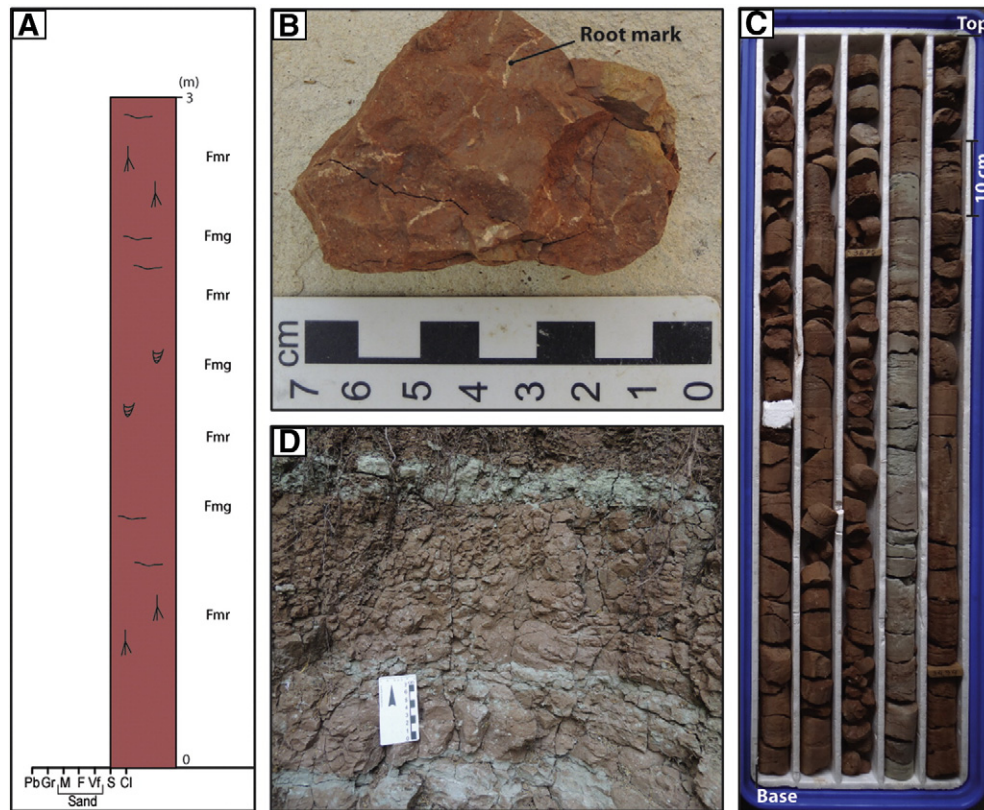


Fig. 12. (A) Typical vertical log of floodplain facies association. (B) Massive, red mudstone (Fmr) with root traces. (C) Thick succession characterized by red (mainly), purple and grayish massive mudstones (Fmr, Fmg) deposited on a well-drained floodplain. (D) Mottled, massive mudstones (Fmr, Fmg) with blocky peds. See Fig. 2 for explanation of symbols used in the sedimentological logs.

The presence of gray to black mudstones, with abundant organic matter, invertebrate shells and plants remains, suggests that these deposits were accumulated and were preserved in an anoxic environment (Paula-Freitas, 2010). The thick and laterally continuous packages of the facies association suggest the presence of a large lake. The presence of stromatolites suggests shallow lacustrine environments (Paula-Freitas, 2010). However, the absence of roots and desiccation features indicates a perennial lake, with no subaerial exposure. The absence of any preserved evaporite facies suggests an abundant supply of fresh water to the lake, where precipitation exceeded evaporation and the concentration of salts was low (Talbot and Allen, 1996). The sandstone beds probably represent infrequent incursion of hyperpynal flows into an inner lake environment that was otherwise characterized by suspension settling of mud. The gravitationally-emplaced sand beds supplied the environment with required nutrients and oxygen, allowing organic activity and bioturbation of the substratum.

6. Sequence stratigraphy framework

In alluvial settings, sequence stratigraphic analysis has focused on the individualization of depositional sequences by the identification and correlation of subaerial unconformities. Internally, the depositional sequences have been subdivided in systems tracts, based on the relative proportion of fluvial channel sand bodies and fine overbank deposits (Shanley and McCabe, 1994). Several studies developed in coastal alluvial systems have subdivided the fluvial depositional sequences in three systems tracts (Lowstand, Transgressive and Highstand system tracts), following a symmetrical cycle of variations in the accommodation/sediment supply (A/S) ratio through time, similar to that observed in marine environments (Shanley and McCabe, 1991, 1994; Wright and Marriot, 1993; Olsen et al., 1995; Van Wagoner, 1995). However, different authors have highlighted the problems of using the traditional

terminology of systems tracts and key surfaces in hinterland continental successions (e.g., Currie, 1997; Dalrymple et al., 1998; Legarreta and Uliana, 1998; Martinsen et al., 1999). Among these different studies, we highlight the work of Martinsen et al. (1999) and subsequently Catuneanu (2006) that proposed a model of fluvial depositional sequences composed of only two systems tracts, formed during a stage of varying rates of positive accommodation: Low-Accommodation and High-Accommodation Systems Tracts. The low-accommodation systems tract (LAST) is characterized mainly by amalgamated channel deposits, in contrast to the high-accommodation systems tract (HAST), formed by overbank or lacustrine-dominated successions. The contact between LAST and HAST is defined by an expansion surface or zone, marked by an increase in the overbank/fluvial channel ratio (Martinsen et al., 1999).

The sequence stratigraphy nomenclature proposed by Martinsen et al. (1999) is the one that best applies to this study, where the fluvial succession is characterized by well-defined cyclic changes in fluvial stacking pattern between amalgamated fluvial channel sandbodies and fine-grained overbank with isolated channels sandbodies or lacustrine intervals. Two fluvial depositional sequences can be identified in the Aptian Barbalha Formation (Fig. 14). Both sequences are composed of low- and high-accommodation systems tracts (sensu Martinsen et al., 1999), recognized and correlated across the studied area.

6.1. Sequence 1

This sequence is at least 60 m thick, the complete section being documented only in one outcrop (Serra da Mãozinha), and in subsurface only the top of the unit was cored (Fig. 14). Sequence 1 is bounded at the base by a sharp and strongly erosional subaerial unconformity that separates fine-grained lacustrine deposits of the Abaiara Formation (top of the Rift Supersequence) and coarse, alluvial sandstones of the Barbalha Formation (base of the Post-rift Supersequence). Sequence 1

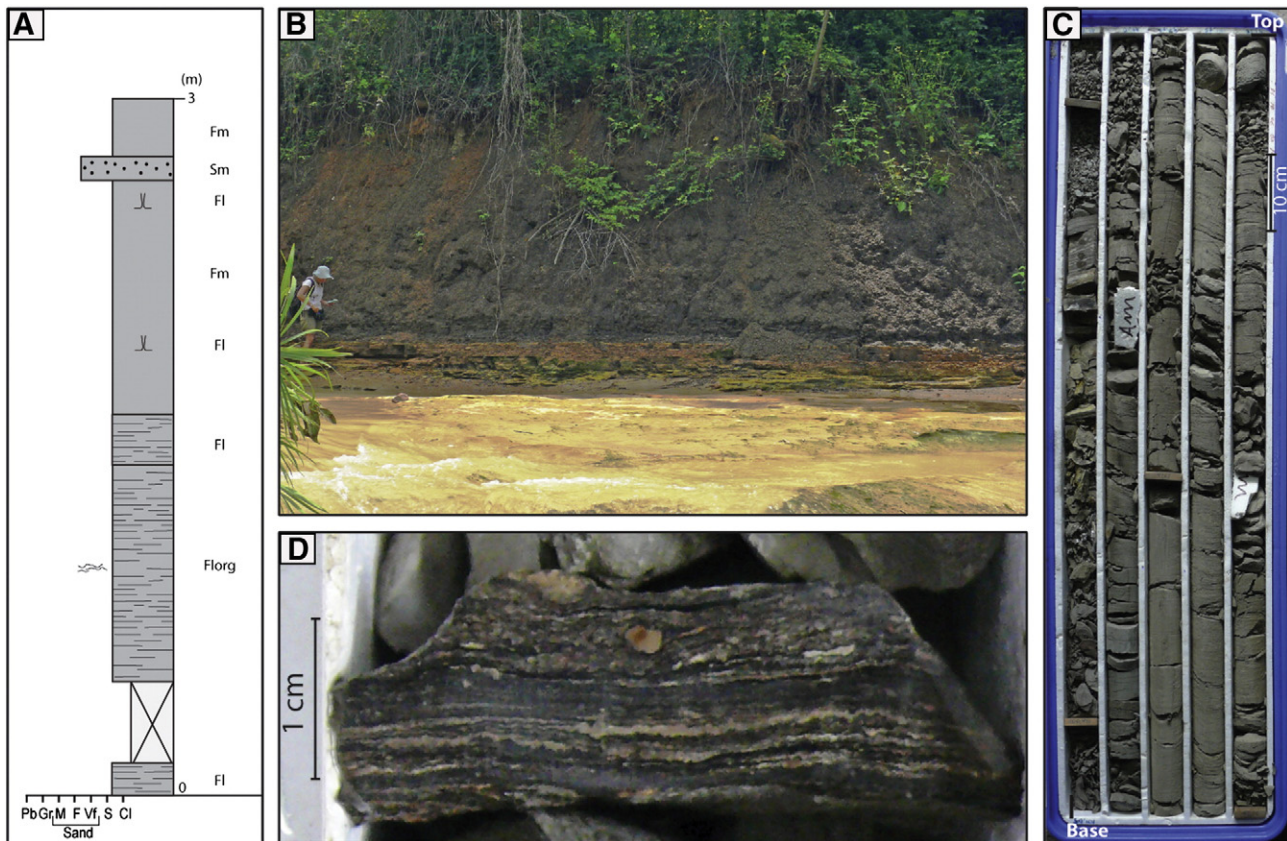


Fig. 13. (A) Typical vertical log of lacustrine facies association. (B) Thick, gray to black, laminated mudstones (Flg). (C) Millimetric to centimetric microbial carbonate laminae or nodules, interlayered with organic-rich, black shales (Florg). See Fig. 2 for explanation of symbols used in the sedimentological logs.

shows a general fining-upward trend. The base of the sequence is characterized by multistorey and multilateral, amalgamated braided fluvial channel sand bodies, interlayered with rare and discontinuous overbank deposits, defining a LAST (Fig. 15A). As a result of facies homogeneity throughout the LAST, recognition and discrimination of individual channel belts sand bodies is uncertain. Only the identification of amalgamated channel complexes is possible. This architectural style suggests that most or all of the alluvial plain was covered by channels, with reduced area to accumulate floodplain deposits. The few floodplain and crevasse splay deposits accumulated in overbank areas were eroded by frequent avulsion of the channels, these deposits having thus a small preservation potential (Miall and Arush, 2001; Leleu et al., 2009). In the middle to upper part of the succession, the amalgamated braided fluvial sand bodies are more frequently interlayered with crevasse splay and floodplain deposits, defining a braided stream system with laterally well-developed overbank areas (Fig. 15B). This depositional architecture suggests a context where the braided channel belt aggrades vertically rather than expands laterally through time (Bentham et al., 1993), indicating a change of the fluvial style within Sequence 1. In this case, erosion of the braided channel belt during avulsion events is not sufficient to remove all overbank material previously accumulated, allowing the preservation of thicker floodplain and crevasse splay deposits. The interval composed of braided fluvial sand bodies interlayered with overbank deposits, is capped abruptly by thin (<10 m thick), laterally persistent lacustrine black shales, pointing to a fast and regional flooding of the basin (Fig. 15C). This lacustrine package is a significant marker in the basin (Batateira layer). The fluvial braided channels with laterally well-developed overbank areas and the lacustrine deposits are interpreted as HAST. The contact between LAST and HAST is transitional, characterized by a

progressive increase in the overbank/channel ratio. Paleocurrent measurements of the fluvial channel deposits throughout Sequence 1 show a mean value to the SE (Fig. 14), indicating that the source area and proximal deposits of Sequence 1 were located to the northwest of the study area.

6.2. Sequence 2

Sequence 2 was recognized in all studied outcrops and wells, with a minimum thickness from 100 to 160 m (Fig. 14). The upper boundary cannot be precisely delineated, although it probably corresponds to the top of the Aptian Crato Member. This sequence is bounded at the base by an erosional subaerial unconformity that can be correlated across the basin. The sequence boundary marked by a subaerial unconformity is placed at the base of extensive, amalgamated fluvial channel bodies that cover the HAST lacustrine deposits at the top of Sequence 1. The basal subaerial unconformity truncated the HAST lacustrine deposits at the top of Sequence 1, with at least more than 50 cm of relief in outcrop. Internally, Sequence 2 shows a general fining-upward trend, being a common occurrence of sandy conglomerates and very coarse-grained sandstones to conglomerates overlying the unconformities (Fig. 14). The lower section comprises amalgamated, multistorey, braided fluvial channel sand bodies, interpreted as LAST (Fig. 15D). The braided fluvial system in Sequence 2 presents morphological features similar to the fluvial model of the LAST in Sequence 1, being characterized by a highly mobile braided fluvial channel belt, marked by unstable banks with low potential of preservation of overbank deposits. The amalgamated fluvial channel sand bodies are overlain by poorly-drained floodplain with fixed fluvial channel deposits, interpreted as an anastomosed fluvial system (Fig. 15E), which are in turn capped by

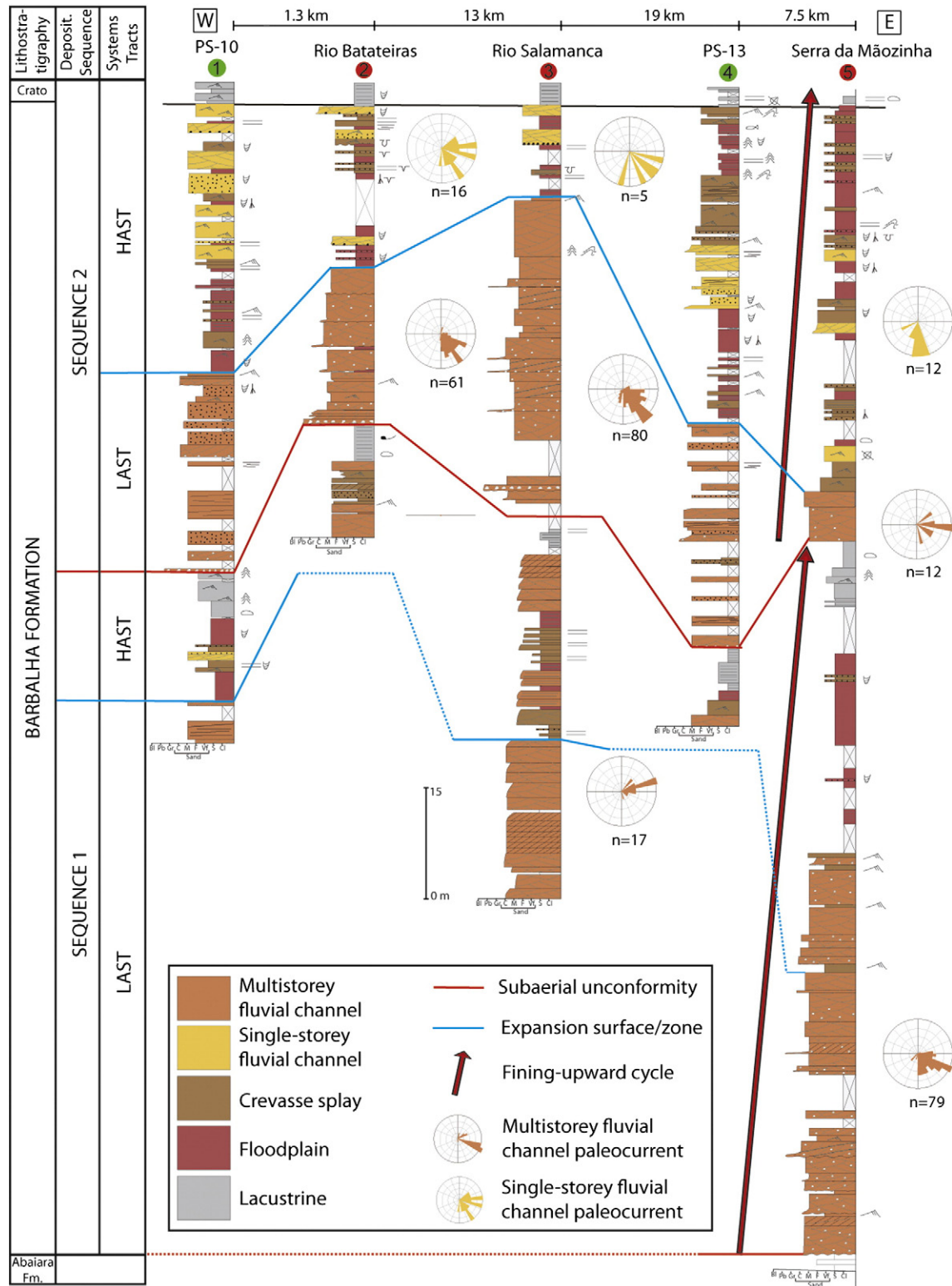
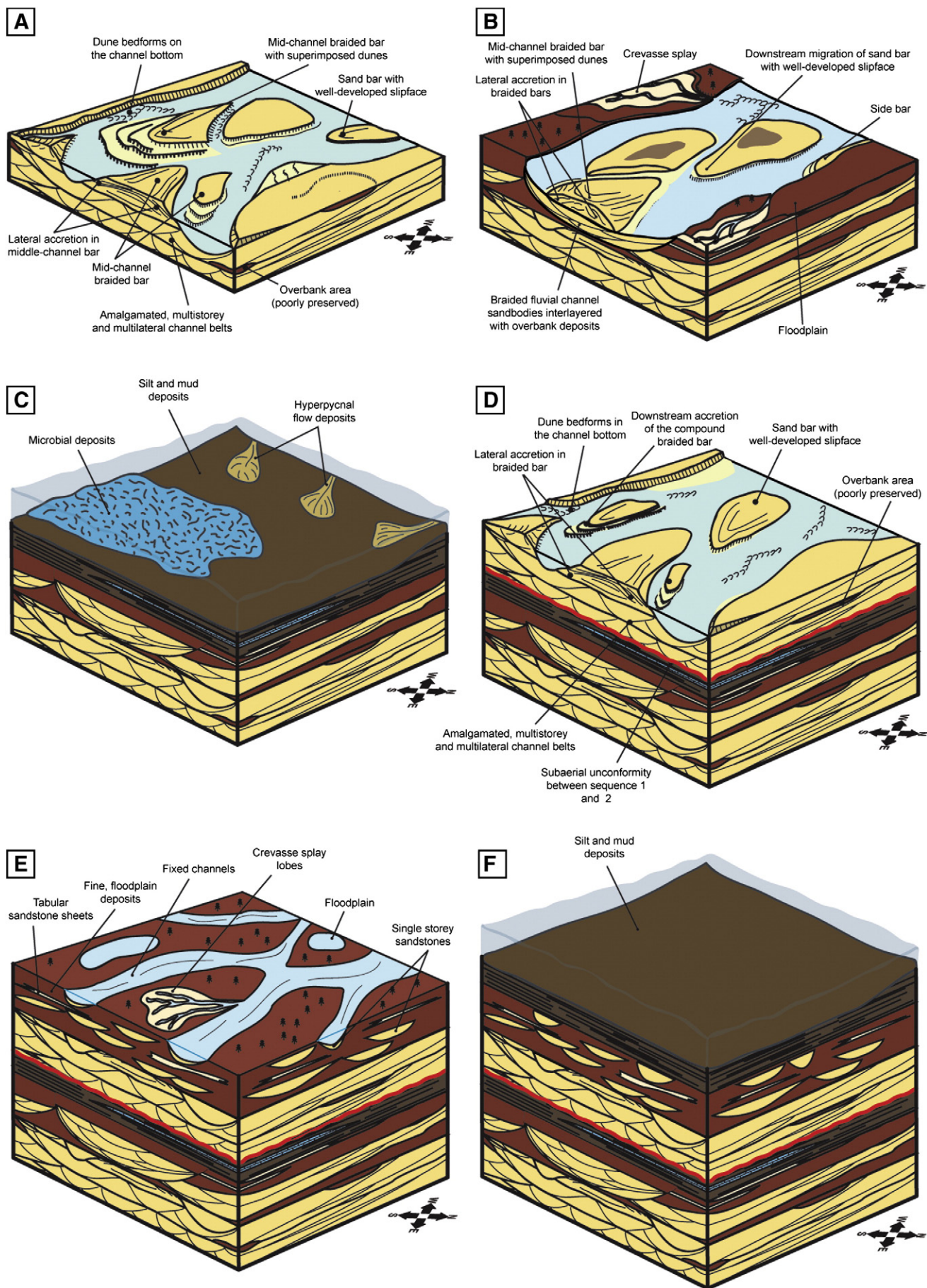


Fig. 14. Regional correlation panel depicting depositional sequences, their bounding surfaces and facies associations (datum: lacustrine deposits at the base of the Crato Member). See Fig. 1 for the location of logs, and Fig. 2 for explanation of symbols used in the sedimentological logs.

lacustrine deposits (Fig. 15F), both grouped in HAST. The contact between LAST and HAST is gradational, marked by a progressive decrease in sand/mud ratio. Paleocurrent measurements of the fluvial channel deposits of Sequence 2 show a consistent pattern to SE, both LAST and HAST, similar to what was described in Sequence 1.

6.3. Change in A/S ratio

The development of systems tracts and bounding surfaces in fluvial systems can be analyzed in terms of the balance between accommodation (A) and sediment supply (S) (Martinsen et al., 1999; Catuneanu,



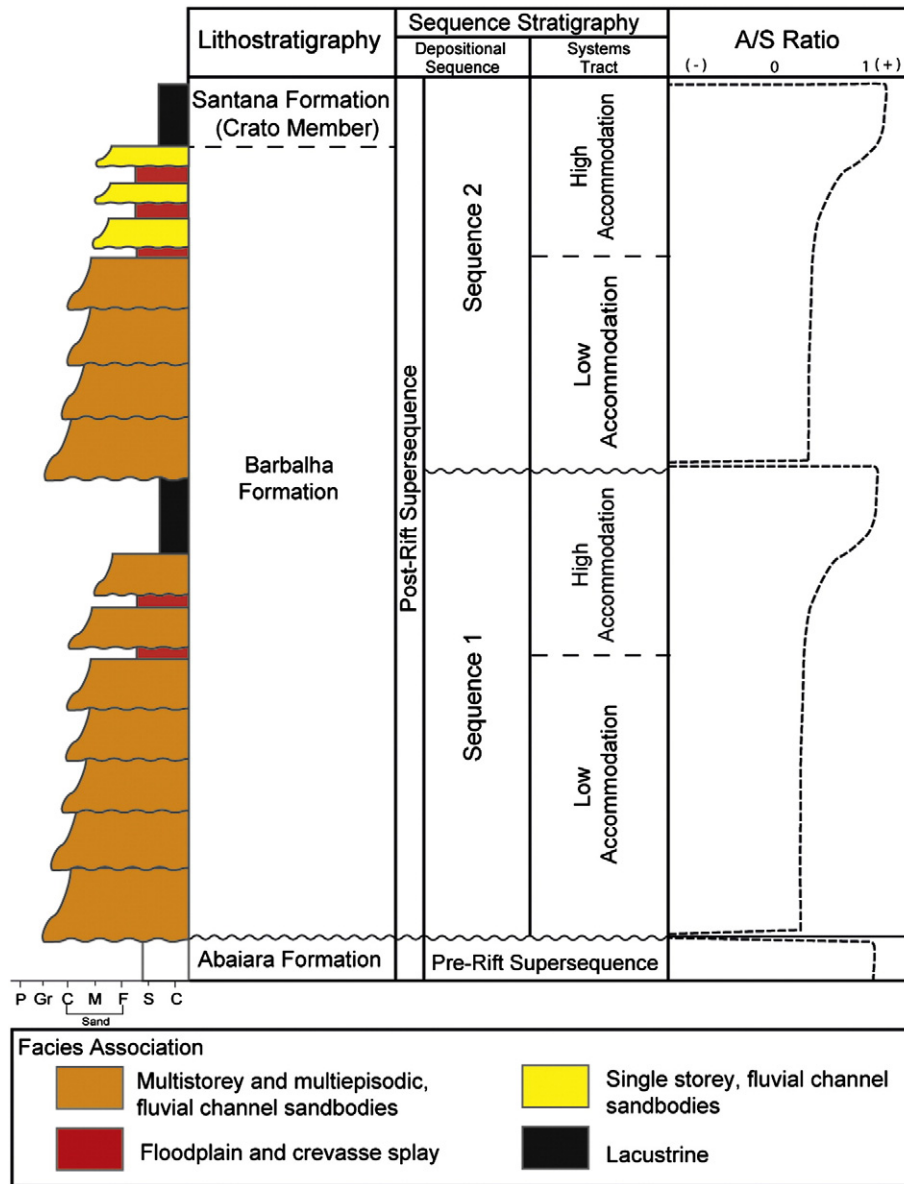


Fig. 16. Sequence stratigraphic model of the fluvial-lacustrine section of the Barbalha Formation. The interval was subdivided into two unconformity-bounded, regional-scale depositional sequences. Each sequence can be divided in low-accommodation and high-accommodation systems tracts. Sediment accumulation takes place during episodes of positive A/S ratio, while unconformities are developed during periods at negative A/S ratios.

2006; Scherer et al., 2007). The A/S ratio is not absolute, since there is no absolute or relative dating which allow the definition of the subsidence and/or sedimentation rate in the basin. The biostratigraphic framework available is of low resolution, not allowing chronostratigraphic refinement. The stratigraphic interval of the Barbalha Formation is contained in one biozone, covering a time interval of about 4 Ma (Coimbra et al., 2002). Thus, the inference about A/S ratio is based on the qualitative fluvial stacking pattern, as discussed by Shanley and McCabe (1994); (386 Martinsen et al. (1999). According these papers, settings with an A/S ratio above 1 indicate a depositional context where the rate of the accommodation generation outpaces the rate of sediment supply, favoring the development of lacustrine environment. If the A/S ratio is between 0 and 1, sediment supply fills the available space, allowing the accumulation of the fluvial systems. Positive A/S ratio (but close to 0) is characterized by multilateral and multiepisodic, amalgamated fluvial channel

sand bodies, while A/S ratios close to 1 are composed of single-storey, ribbon or sheet fluvial channel sand bodies encased within fine-grained overbank deposits. Finally, where the A/S ratio is negative, sediment bypass occurs with formation of the regional-scale erosion surfaces (subaerial unconformity). A general sequence stratigraphic model of the Aptian Barbalha Formation fluvial succession is shown in Fig. 16.

The unconformities at the base of sequences 1 and 2 record a negative A/S ratio. The subsequent periods of accumulation of sequences 1 and 2 are related with a rise of stratigraphic base level, indicating a positive A/S ratio. The dominance of fluvial channel belt sandstones relative to fine-grained, overbank deposits at the base of sequences 1 and 2 gives an A/S ratio positive but close to 0, defining a low-accommodation systems tract where the addition of the new accommodation was very slow. The vertical increase in the occurrence of the overbank deposits

Fig. 15. Summary depositional models for the different systems tracts and depositional sequences of the Aptian succession (Barbalha Formation) in the Araripe Basin. (A) LAST of Sequence 1. (B) Base of HAST of Sequence 1. (C) Top of HAST of sequence 1. (D) LAST of Sequence 2. (E) Base of HAST of Sequence 2. (F) Top of the HAST of Sequence 2 in the studied area.

interlayered with braided (sequence 1) and anastomosing fluvial channel (sequence 2) sand bodies suggests an increase in accommodation and a change in the A/S ratio to values near 1, indicating the development of a high-accommodation systems tract. The abrupt occurrence of lacustrine deposits overlying fluvial systems in a large area at the top of high-accommodation systems tract of the sequence 1 and 2 implies a marked increase in the A/S ratio to values above 1.

7. Controls on sedimentation

The stratigraphic architecture of alluvial successions can be controlled by three allogenic processes (relative sea level, climate and tectonics). However, it is extremely difficult to discriminate correctly which specific factor is controlling fluvial accumulation in a basin. Different authors have highlighted the influence of relative sea level change in the stratigraphic architecture of fluvial successions (e.g., Shanley and McCabe, 1991; Aitken and Flint, 1995; Van Wagoner, 1995; Fanti and Catuneanu, 2010; Corbett et al., 2011; Benvenuti and Del Conte, 2013). Nevertheless, numerous studies have demonstrated that a fluvial equilibrium profile is influenced by relative sea level only in regions close to the coast (e.g., Shanley and McCabe, 1991, 1994; Schumm, 1993; Aslan and Autin, 1999; Blum and Tornqvist, 2000; Plint et al., 2001; Posamentier, 2001; Amorosi and Colalongo, 2005; Catuneanu, 2006; Catuneanu et al., 2011). The distance toward the continent where sea level influences the behavior of fluvial equilibrium profile depends on the river gradient and discharge. Blum and Tornqvist (2000) suggested that the distances can be 300–400 km for low gradient streams with a high sediment supply, and less than 40 km for streams with a steep gradient and a low sediment supply. The Barbalha Formation was deposited in a continental context, without sedimentological and paleontological evidence of marine influence. All fossils collected in this unit indicate low-salinity lacustrine bodies, such as non-marine ostracods (Farina, 1974; Lima and Perinotto, 1984; Hashimoto et al., 1987; Coimbra et al., 2002; Rios-Netto et al., 2012). Marine microfossils such as dinoflagellates (*Spiniferites seghiris* and *Subtilisphaera* sp.; Coimbra et al., 2002) occur only in the Santana Formation (Crato and Romualdo Members), which are stratigraphically above the Barbalha Formation (Coimbra et al., 2002).

Different authors have highlighted climate as an important control on large-scale stratigraphic architecture through induced changes in sediment supply that hold the fluvial equilibrium profile. However, sedimentological data indicate a climatic uniformity along the deposition Barbalha Formation. The LAST deposits of sequences 1 and 2 are characterized by fluvial channel sand bodies composed of mainly side and middle bars, compatible with deep and perennial braided rivers. Additionally, the abundance of lower flow regime structures (trough and planar cross bedding) and the almost absence of the transcritical and supercritical structures (low-angle cross stratification and horizontal laminations) reinforce the interpretation of these deposits as perennial river systems (Allen et al., 2013). However, the presence of reactivation surfaces regularly spaced in macroforms indicates that, although perennial, the river was characterized by frequent variations in discharge. The HAST fluvial and lacustrine deposits of sequences 1 and 2 indicate similar climatic conditions. The braided rivers of the HAST in Sequence 1 is also dominated by cross-bedding and downstream accretion macroforms, suggesting that the flow inside the channel was perennial, although the presence of reactivation surfaces indicates some degree of discharge variation. The anastomosing river channels of the HAST in Sequence 2 are dominated by structures formed under lower flow regimes, suggesting that fluvial channels experienced a perennial flow (Miall, 1996; Allen et al., 2013). However, overbank deposits interlayered with sand bodies within fluvial channels show evidence of frequent variations in river discharge. The intercalation of poorly- and well-drained floodplain deposits suggests variation in the water table in the floodplain area. The abundant presence of crevasse deposits composed of multiple episodes of flooding, associated with the occurrence of subaerial exposure features (mud cracks and paleosols), indicates frequent events of

fluvial channel overflow and drowning of the overbank areas, followed by periods of decrease of river discharge and development of well-drained floodplains. The lacustrine deposits at the top of the HAST in turn are characterized by organic mudstones, with no evidence of subaerial exposure, which points to a perennial nature of the lakes. However, the presence of microbial carbonate laminae and low-diversity ostracod assemblages interlayered with mudstones indicates stressful environments and/or periods, possibly associated with high-frequency variations in temperature, and/or degree of oxygenation of the water.

The sedimentological evidence indicates that, despite variations in river discharge, and temperature and/or oxygenation of the lakes, the climate was relatively humid or subhumid, without major climatic changes during the deposition of sequences 1 and 2. Thus, tectonics must be the main control of changes in the A/S ratio. The abrupt contact between fluvial and lacustrine deposits can be taken as evidence for tectonics as the main factor controlling the depositional architecture. Although the degree of amalgamation of fluvial channels gradually decreases toward the top of sequences, the transition between fluvial and lacustrine systems is abrupt, suggesting major changes in the basin sedimentation regime and sediment supply from the source area, compatible with tectonically-driven fluvial successions (e.g., Plinth and Browne, 1994; Martinsen et al., 1999; Fanti and Catuneanu, 2010; Allen et al., 2014). Climatic changes tend to produce gradual changes in the stratigraphic succession (Allen et al., 2013). The rapid transition of fluvial to lacustrine deposits can be better explained by a rapid increase in subsidence, such as already interpreted for other fluvial units, such as Boss Fort Formation, Cumberland Basin, Canada (Plinth and Browne, 1994) and Campanian Erikson Sandstones, Western Interior Basin of the USA (Martinsen et al., 1999).

Tectonic movements controlled the A/S ratio during accumulation of the fluvial–lacustrine Barbalha Formation. Periods of negative A/S ratio, responsible for the generation of unconformities, can be linked to tectonic uplift of the cratonic area. In turn, periods of positive A/S ratio would be linked to subsidence in the basin. The LAST represents a low subsidence context, whereas the HAST is related to widespread increase in subsidence. Nevertheless, a syn-depositional fault system was not identified in the Barbalha Formation, which suggests that tectonics acted primarily through epeirogenic regional movements. Regional variations in the regional stress fields acting in the lithosphere are capable of producing vertical movements on sedimentary basin margin, enlarging or reducing the amplitude of the resulting flexural deformation (Cloetingh et al., 1985; Cloetingh, 1988; Cloetingh and Kooi, 1992; van Balen et al., 1998). This type of tectonic movement is compatible with the tectonic model of the basin, linked to a wide sag basin developed in a post-rift stage, after the cessation of extensional movements related to Gondwanaland breakup (Ponte and Appi, 1990; Chang et al., 1992). The mechanism of subsidence of this phase is dominantly thermal, and mechanical subsidence associated with fault movement is quite restricted, with little influence on the creation of accommodation space. The epeirogenic movements generated few changes in depocenters from one sequence to another, as evidenced by measurements of paleocurrent of sequences 1 and 2, which have a constant paleoflow to the SE.

8. Conclusions

1. The Barbalha Formation records deposition within a fluvial and lacustrine environment accumulated in a sag basin developed during the post-rift phase. This stratigraphic unit encompasses five main facies associations ascribed to: (i) multistorey, amalgamated fluvial channel sand bodies, (ii) single-storey fluvial channel sand bodies, (iii) crevasse splay, (iv) floodplain, and (v) lacustrine deposits.
2. Two depositional sequence bounded by regional erosional unconformities are proposed. Both sequences shows a general fining-upward trend, characterized by amalgamated channel-fill beds at the base (low-accommodation systems tract) that pass

upwards into floodplain-dominated and lacustrine successions (high-accommodation systems tract). The recognized sequences can be correlated across more than 100 km, thereby establishing a regional sequence-stratigraphic framework. The stratigraphic framework proposed is similar to other hinterland sequence models, such as those by Martinsen et al. (1999) for the Erikson Sandstone (Campanian), USA.

3. The development of depositional sequences in the Barbalha Formation reflects variations in the A/S ratio. Sequence boundaries (subaerial unconformities) formed during periods of negative A/S ratios, while accumulation periods are associated with positive A/S ratios. The low-accommodation systems tracts are characterized by positive (close to zero) A/S ratios, while the high-accommodation systems tracts are associated with A/S ratio value near from 1 (fixed and isolated channels within overbank deposits) or greater than 1 (lacustrine deposits). Variations in A/S ratio are probably related to tectonic subsidence and uplift of the basin.

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